

MedGate-FL: Capability-Isolated Federated Adapters for Tiered Medical AI Access

[Preliminary technical report — synthetic validation only, real-data tier license-gated]

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Draft of August 26, 2026 — Phases 0–5 complete on two synthetic fixtures (null-signal and hierarchical); real-data tier license-gated

Abstract

Federated learning is often described as protecting patient privacy because raw data never leaves an institution, but this protects data *locality*, not the gradients, representations, or final model that federation produces. We study a distinct, narrower problem: whether a federated medical model can expose a useful *public*, *coarse-grained* diagnostic capability while keeping a *fine-grained* diagnostic capability restricted to authorized users, and whether that restriction survives realistic attacks rather than merely hiding a prediction head. We call this *capability isolation*. We implement MedGate-FL, a federated model with a shared backbone, a public coarse head, and a restricted fine-grained LoRA adapter, trained under six objectives (coarse-only, hidden-fine-head, adapter-isolation, adversarial suppression, orthogonality, and their combination), and evaluate it against a *fair* baseline family – a random-frozen-LoRA negative control, a coarse-pretrained FedLoRA baseline, an ImageNet-pretrained FedLoRA baseline, and a full-fine-tune upper bound – under four attacker models (honest-but-curious server, unauthorized model recipient, black-box query access, and colluding/malicious clients) with precisely named attacks, including a genuine parameter-space adapter-recovery attack distinct from the fresh-training attacks. We also implement a conservative cryptographic layer (AES-GCM adapter-at-rest encryption, a minimal centralized-IAM authorization service, backed by a negative-test suite covering nonce/key uniqueness, ciphertext tampering, and AAD binding), a pairwise-masking secure-aggregation simulation whose correctness *and* failure modes (client dropout, seed collusion) are both directly demonstrated rather than only asserted, DP-SGD via Opacus with monotonicity-tested accounting, and five revocation/unlearning methods compared against full retraining as the gold standard, scored with a corrected, direction-invariant membership-inference metric (SymmetricAUC / AttackAdvantage) after a review caught the previous formula’s sign bug. **The primary dataset, FLamby Fed-ISIC2019, requires a human to accept its license before download (§6);** that step has not yet been completed, so every result in this draft is run on synthetic fixtures, not real medical images. We use two: a *null-signal fixture* that mirrors Fed-ISIC2019’s shape (6 centers, 8 classes) but carries no real diagnostic signal, useful only as a pipeline smoke test; and a new *hierarchical fixture*, built from Fed-ISIC2019’s own published per-class institution counts, with an analytically constructed coarse/fine label hierarchy and per-institution/per-patient nuisance variation, which we independently verify is learnable by both the coarse and fine heads before using it for any capability-isolation claim. **Every empirical number in this paper is therefore a validation result on synthetic data, not a scientific finding about capability isolation on real medical images,** and is labeled as such throughout. **[RESULT PENDING: real Fed-ISIC2019 results for every table in §7–§9].** On the hierarchical fixture, the fair pretrained baselines show a real, non-trivial coarse/fine capability gap that the null-signal fixture cannot exhibit – but capability isolation does *not* yet show a clean win over that gap: at the compute budgets we could afford, the best disjoint probe’s representation leakage is comparable to or exceeds authorized fine-task

utility for most isolation objectives, a finding we report rather than suppress (§7). What is established here: a tested, reproducible implementation of the experimental pipeline on two synthetic fixtures (data, fair baselines, six training objectives, a renamed and extended attack suite, four privacy arms, five unlearning methods, statistical aggregation, figure/table generation, and this document); several genuine engineering findings (two Opacus integration bugs, one flawed confidentiality test, and one buggy membership-inference formula, each found, fixed, and documented rather than quietly patched); one class-level unlearning confound identified and corrected with an independent control population, with the original confounded numbers preserved only in a labeled appendix; and the negative capability-isolation finding above. Whether capability isolation actually works on real medical images is, at this point, unknown.

1 Introduction

Cross-silo federated learning lets multiple hospitals train a shared model without centralizing patient data [McMahan et al., 2017]. This is frequently summarized as “federated learning protects privacy,” but the protection is narrower than that: only the raw training images stay local. The gradients or weight updates each institution sends are still exposed to the aggregating party [Zhu et al., 2019]; the final trained model, once distributed, is exposed to whoever receives it; and nothing about the FedAvg protocol itself limits what that model can be made to reveal under later probing, fine-tuning, or querying. Prior work on medical federated learning [*background source, not an empirical claim of this paper* – Ogier du Terrail et al. [2022]] and on securing federated training generally treats three problems separately: protecting the training updates (secure aggregation [Bonawitz et al., 2017], differential privacy [Abadi et al., 2016]), protecting model intellectual property after distribution, and authorizing who may query or fine-tune a deployed model.

This paper studies a fourth, distinct problem that sits underneath all three: a single federated model may need to expose a coarse, low-risk capability to a broad audience while keeping a fine-grained, higher-risk diagnostic capability restricted to an authorized tier. A concrete instance, and the one this paper builds around, is skin-lesion classification: a coarse melanocytic/keratinocytic/other signal might be safe to expose broadly, while the eight-way fine diagnosis (including melanoma vs. benign nevus) is exactly the output a clinical safety or liability framework would want gated. We call keeping that fine capability restricted, while a coarse capability remains useful, **capability isolation**.

The naive approach – train both heads, only expose the coarse one – is not capability isolation, because the fine information may still be linearly or near-linearly recoverable from the shared backbone’s representation even though the fine head itself is hidden (this is exactly the `hidden_fine_head` arm we implement and probe in §9). Establishing whether that residual leakage is small in practice, under realistic attacker budgets, is the empirical core of this paper: it is not assumed, it is measured, with a battery of representation probes (linear, nonlinear, k -NN, few-shot) and reconstruction/extraction attacks (§8).

Contributions. (1) A capability-isolation training objective and architecture (§5) compared against five simpler alternatives under matched compute, not assumed superior. (2) An attacker-explicit evaluation across four threat models (§4) with attacker knowledge, access, and budget recorded for every attack result. (3) A conservative cryptographic/authorization layer (AES-GCM, centralized IAM, never RSA-encrypting tensors directly) and a from-scratch secure-aggregation simulation, with an honestly reported instance of building the wrong test for a security property and correcting it (§7.3). (4) A revocation-vs-unlearning study that operationalizes, rather than asserts, the project’s central non-claim that revoking a key does not remove a model’s already-learned

influence (§10). (5) A fully reproducible, open pipeline (§14) in which every number in this draft traces to a config, seed, git commit, and raw JSON file – currently run only on a synthetic fixture, pending a human license-acceptance step for the real data (§6).

2 Background and Motivation

Medical imaging tasks routinely have a natural coarse/fine label hierarchy: a triage-level category (e.g., “possibly malignant, refer”) and a diagnosis-level category (e.g., a specific carcinoma subtype). Access to the fine category carries more clinical and liability weight, and in many real deployments would be gated to credentialed users even when a coarse signal is offered as a lower-stakes public or semi-public service (e.g., a consumer-facing triage tool). Federated learning is attractive for training such a model because the fine-grained labels, sourced from multiple hospitals, cannot be centralized – but federating the *training* says nothing about how to gate the *resulting model’s* capability once training finishes. That is the gap this paper works in.

3 Related Work

Verification status. Every citation in this section is drawn only from the VERIFIED rows of docs/literature_matrix.csv (16 of 59 seed-bibliography entries as of this draft); the other 43 are deliberately not cited here even where the project brief’s seed bibliography names them, pending a primary-source check (docs/literature_matrix.md records exactly which and why). Sections below that would normally cite an unverified source instead say so explicitly with **[UNVERIFIED:]**.

Federated learning. FedAvg [McMahan et al., 2017] established the communication-efficient local-SGD-then-average pattern this paper’s baselines and proposed method both build on. FedProx [Li et al., 2020] adds a proximal term for statistical/systems heterogeneity; SCAFFOLD [Karimireddy et al., 2020] corrects client drift with control variates; FedAdam and related adaptive federated optimizers [Reddi et al., 2021] replace plain averaging with a federated adaptive step. Kairouz et al. [2021] survey the broader space. **Gap:** none of these methods, or the survey, address exposing part of a trained model’s capability while restricting another part – they optimize *how* a single, uniformly-accessible model is trained, not *what parts of it different recipients are allowed to use*.

Cross-silo medical federated learning and benchmarks. FLamby [Ogier du Terrail et al., 2022] is the direct dataset and baseline source for this paper (§6); it is a benchmark and dataset paper, not a security paper, and makes no capability-isolation claim. **[UNVERIFIED: Sheller et al. 2020; Rieke et al. 2020; Teo et al. 2024 – broader medical-FL feasibility/survey claims are not cited pending verification.]** **Gap:** realistic cross-silo benchmarks like FLamby establish that federated training is feasible on natural institutional splits; they do not evaluate what a distributed model leaks about a restricted sub-task.

Parameter-efficient adaptation and model protection. LoRA [Hu et al., 2022] is the adapter mechanism A_ϕ in MedGate-FL’s architecture (§5). LoREnc [Ahn et al., 2026] proposes a training-free spectral-truncation/orthogonal-reparameterization scheme so an unauthorized holder of a LoRA adapter gets structurally degraded outputs while an authorized one recovers full performance – conceptually the same goal as this paper’s encrypted-adapter-at-rest design, but LoREnc’s

mechanism is a transform on the adapter weights themselves (obfuscation), whereas this paper’s crypto layer (§7.3) uses conventional AEAD encryption and evaluates the resulting model separately for representation leakage (§9) – the two are complementary, not the same claim, and this paper does not implement or evaluate LoREnc’s specific transform. **[UNVERIFIED: FedShield-LLM (Mia and Amini 2025); Encryption-Friendly LLM Architecture (Rho et al. 2024); FL-DABE-BC (Narkedimilli et al. 2024) – not cited pending verification.]** **Gap:** LoRA and LoREnc each solve one piece (parameter efficiency; adapter obfuscation) in isolation from the federated, multi-attacker evaluation this paper runs them through.

Secure aggregation and differential privacy. Practical secure aggregation [Bonawitz et al., 2017] hides individual client updates from the server via pairwise masking; this paper’s secure-aggregation module (§7.3) simulates exactly that mathematical core in a single process, explicitly not the full dropout-robust protocol. DP-SGD [Abadi et al., 2016] bounds what a trained model’s parameters can reveal about any single training example via gradient clipping and noise; this paper applies it per-client via Opacus. Client-level DP [Geyer et al., 2017, McMahan et al., 2018] and the foundational DP definitions [Dwork and Roth, 2014] inform the framing but are not themselves implemented as separate baselines here – Phase 4 (§7.3) applies record-level DP-SGD per client, not the client-level variant these two papers propose; that distinction is a concrete, not-yet-implemented extension (§12). **Gap:** neither mechanism, by itself or combined (as we test directly, §7.3), is a capability-isolation mechanism – both protect *how training happened*, not *what the resulting model exposes to different recipients*, and this paper does not claim secure aggregation prevents the membership or property inference it separately evaluates in §8.

Inference and extraction attacks. Deep Leakage from Gradients [Zhu et al., 2019] and its refinements iDLG [Zhao et al., 2020] and gradient inversion via cosine-similarity matching [Geiping et al., 2020] reconstruct a training input from its raw gradient; membership inference [Shokri et al., 2017] and its comprehensive white-box extension [Nasr et al., 2019] distinguish training members from non-members via model behavior; property inference [Melis et al., 2019] infers a property of a participant’s training set (not membership of a specific record) from collaborative-learning updates; model inversion [Fredrikson et al., 2015] and GAN-based active leakage [Hitaj et al., 2017] reconstruct class-representative inputs rather than specific records; model extraction [Tramèr et al., 2016] and training- data extraction from language models [Carlini et al., 2021] clone or recover content from black-box query access. A simplified known-label variant of DLG, loss-threshold membership inference, model extraction (as fixed-budget hard-label distillation and auxiliary-data adapter fine-tuning recovery), and property inference [Melis et al., 2019] are all implemented and evaluated in §8 against MedGate-FL specifically, adapted to ask whether the *public* path leaks the *restricted* task – a different question from these attacks’ original centralized or generically-federated framing; property inference in particular is evaluated leave-one-client-out in §8’s A4 integrity subsection, with the small-client-count caveat stated there. iDLG, the cosine-similarity gradient-inversion variant, comprehensive white-box membership inference, model inversion, and GAN-based leakage remain cited as background here but are not implemented as additional attacks (§12).

Federated and medical unlearning. Machine unlearning’s exact gold standard, retraining from scratch, and cheaper approximations were formalized by Ginart et al. [2019] (data deletion, formal guarantees) and Bourtole et al. [2021] (SISA training for efficient exact unlearning) outside the federated setting; this paper’s `full_retrain` (§10) is exactly the gold standard both papers

take as their baseline. In the federated setting, [Wu et al. \[2022\]](#) erase a client’s contribution via knowledge distillation, [Halimi et al. \[2022\]](#) via local unlearning followed by a few federated recovery rounds, and [Zhang et al. \[2023\]](#) via a differentially-private FedRecovery mechanism; [Deng et al. \[2024\]](#) give the first federated-unlearning framework specifically for medical imaging (intracranial hemorrhage and skin-lesion diagnosis). None of these four methods is reproduced here – this paper’s `gradient_ascent_unlearning` (§10) is a generic, simpler technique (ascend on the removed data’s loss, then recover on retained data), deliberately not claimed as an implementation of any of them, though implementing one of them (Halimi et al.’s local-unlearning-then-recovery approach is architecturally the closest fit to this paper’s federated setup) is a natural next step (§12). [Chen and Shu \[2026\]](#) (Lethe) benchmark twelve federated-unlearning methods across eight medical-imaging tasks and multiple removal granularities, finding that the *difficulty of the forgetting request*, not the unlearning method, dominates performance differences, and that client-level removal often has minimal utility impact – making residual membership signal, not utility loss, the metric that actually distinguishes methods. This is the closest prior benchmark to this paper’s Phase 5, and its finding directly motivates measuring forgetting via a membership-inference proxy rather than utility drop alone, which is what we do. **Gap:** none of the federated-unlearning papers above evaluates unlearning against a capability-isolated architecture, where an “adapter deletion” operation is available that has no counterpart in a monolithic model, and none is medical-imaging-specific except [Deng et al. \[2024\]](#), which does not study capability isolation either.

Access control and cryptography, and blockchain-assisted FL. Ciphertext-policy [[Bethencourt et al., 2007](#)] and key-policy [[Goyal et al., 2006](#)] attribute-based encryption let a policy over attributes (rather than a single key) gate decryption – a natural fit for “authorized if credentialed as X” access rules, but not implemented here: this paper’s crypto layer (§7.3) uses conventional AES-GCM with a centralized-IAM key registry, which is simpler and matches the project’s cryptographic-conservatism rule, at the cost of not supporting CP-ABE/KP-ABE’s richer policy expressiveness. Fully homomorphic encryption [[Gentry, 2009](#)] and its practical approximate-arithmetic descendant CKKS [[Cheon et al., 2017](#)] would let computation happen directly on encrypted adapters or activations; explicitly out of scope on this project’s CPU-only hardware (Table 2), stated as such rather than attempted at a scale that couldn’t be evaluated honestly. **[UNVERIFIED: Every blockchain-FL systematic-review source the project brief names remains unverified as of this draft; see docs/literature_matrix.md.]** **Gap:** none of the ABE or FHE constructions above has been evaluated, in their own papers, against this project’s specific four-adversary threat model (§4) or a federated capability-isolation architecture; this paper’s contribution is that evaluation, not a new construction. Blockchain-assisted authorization (project brief Phase 8) is not yet implemented; see §12.

4 Problem Definition and Threat Model

4.1 Capability isolation, formally

Let f_θ be a shared backbone, h_c a public coarse head, A_ϕ a restricted adapter, and h_f a restricted fine head. The public path is $\hat{y}_c = h_c(f_\theta(x))$; the authorized fine path is $\hat{y}_f = h_f(f_\theta(x) + A_\phi(f_\theta(x)))$. Capability isolation holds, for a given utility metric U and attack budget, when (a) $U(\hat{y}_c)$ is close to a plain federated coarse baseline, (b) $U(\hat{y}_f)$ for an authorized user is close to a plain (unprotected) federated adapter’s fine utility, and (c) no attacker in Table 1 within its stated budget recovers fine utility much above what is recoverable from the public output alone. This is an empirical claim, evaluated per attacker below, not an architectural guarantee.

4.2 Threat model

Table 1: Threat model. Every attack in §8 states its exact knowledge/access/budget in its own result record, not only here.

ID	Adversary	Access	Goal
A1	Honest-but-curious aggregation server	Permitted protocol outputs (raw client updates under plaintext FedAvg; only the sum under secure aggregation)	Infer client data or client properties
A2	Unauthorized model recipient	Public backbone + coarse head + encrypted/transformed restricted adapter + public auxiliary data + normal compute	Recover fine capability without the decryption key
A3	Black-box unauthorized user	Query access to an authorized endpoint, fixed query budget	Distill / extract fine capability
A4	Malicious/colluding clients	Normal client role, one or more rounds	Poison, backdoor, replace the model, or infer other clients' properties

4.3 Claims and non-claims

This paper’s claims are restricted to what §7–§10 actually measure, using *capability isolation*, *restricted capability*, *empirical attack resistance*, and *simulated cross-silo evaluation* in the precise senses just given. It explicitly does **not** claim: that federated learning inherently guarantees privacy; that a public benchmark dataset proves compliance with medical privacy law; that an encrypted adapter stops an attacker from independently training a new model on public data; that the LoREnc-style or AES-GCM mechanisms discussed provide security beyond what is stated for them; that deleting a key removes a patient’s learned influence (§10 measures this directly); that simulated clients on one workstation equal a real multi-hospital deployment; or that any model here is suitable for clinical diagnosis. A restricted model that confidently emits a *wrong* fine-grained prediction is not treated as a successful defense anywhere in this paper – see the calibration/abstention discussion in §12.

5 MedGate-FL

5.1 Architecture

[*implemented, tested*] `medgate/models/backbone.py` implements exactly the components in §4: a small CNN backbone f_θ (sized for the CPU-only hardware in Table 2, not a claim about production architecture), a linear coarse head h_c , a zero-initialized LoRA-style adapter A_ϕ (rank r , matching standard LoRA initialization [Hu et al., 2022] so an untrained adapter is a no-op residual), a linear fine head h_f , and an always-present adversary head used only by the adversarial/combined objectives below (§5.2). Every one of the six training objectives compared in §9 uses this identical architecture and differs only in loss function, so the ablation isolates the objective, not model capacity.

5.2 Training objectives

[implemented, tested] `medgate/federated/capability_isolation.py` implements, and §9 compares without assuming any is superior:

- **coarse_only**: $\mathcal{L}_{\text{coarse}}$ alone (the fine task is never trained at all).
- **hidden_fine_head**: $\mathcal{L}_{\text{coarse}} + \lambda_f \mathcal{L}_{\text{fine}}$, but the fine head reads $f_\theta(x)$ directly (no adapter) – the naive “just don’t expose the head” baseline.
- **adapter_isolation**: the architecture in §5’s full path, joint coarse+fine loss, no extra terms – the reference “plain adapter” point ARR is measured against.
- **adversarial**: adapter_isolation plus a gradient-reversal adversary trying to predict the fine label from the *public* representation $f_\theta(x)$.
- **orthogonal**: adapter_isolation plus a penalty on the squared cosine similarity between $f_\theta(x)$ and the adapter’s residual contribution $A_\phi(f_\theta(x))$ – one candidate operationalization of $\mathcal{L}_{\text{orth}}$, not claimed to be the only or best one.
- **combined**: adversarial + orthogonal together – the full proposed objective, evaluated exactly as skeptically as the other five.

6 Experimental Methodology

6.1 Hardware

[measured] This is the binding constraint on every design decision below: model width, batch size,

Table 2: Measured hardware, captured 2026-08-24 via `scripts/report_hardware.py` (`docs/hardware_report.md`).

CPU	8 logical cores, no GPU (<code>nvidia-smi</code> absent)
RAM	7.4 GiB total, ~ 3.3 GiB available at capture time
Disk	55.7 GiB free
Software	Python 3.12.3, PyTorch 2.13.0 (CPU build), Opacus 1.6.0

and which attacks/phases are attempted at all (`docs/execution_plan.md` states, per phase, what is explicitly out of scope on this hardware – FeTS/BraTS, ADNI, full FHE, and any multi-node blockchain network).

6.2 Primary dataset: FLamby Fed-ISIC2019

[verified facts; download *BLOCKED-LICENSE*] Verified directly against the official owkin/FLamby repository and the ISIC Challenge data page on 2026-08-25 (full source list in `docs/literature_matrix.csv`, evidence in `docs/evidence_ledger.csv`):

- 23,247 skin-lesion images with identifiable source center, across 6 centers (BCN, HAM_vidir_molemax, HAM_vidir_modern, HAM_rosendahl, MSK, HAM_vienna_dias).
- Eight fine-grained classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC.
- A predetermined, center-stratified train/test split.
- License: CC BY-NC 4.0 for ISIC2019 imagery; the HAM10000-sourced portion carries separate Harvard Dataverse terms and requires citing Tschandl et al. [2018].
- Official pooled baseline hyperparameters (from FLamby’s own `common.py`): 6 clients, batch size 64, Adam at $\text{lr } 5 \times 10^{-4}$, 20 pooled epochs – recorded so this paper’s matched-budget

baselines have an official reference point, even though this paper’s own budgets (Table 3) differ (dictated by the hardware in Table 2, not by FLamby’s GPU-scale defaults).

Downloading the images requires a human to accept the ISIC2019 and HAM10000 licenses in a browser (exact procedure: `scripts/download_fed_isic2019_INSTRUCTIONS.md`); this project does not and will not automate that step. As of this draft it has not been completed, so **every result in §7–§9 below runs on a synthetic fixture instead (§6.4)**. **[RESULT PENDING: all Fed-ISIC2019 real-data tables.]**

6.3 Fed-IXITiny (planned extension)

[verified facts; not yet run, Phase 7] ~566 T1-weighted MRI subjects, 3 London hospitals (Guy’s, Hammersmith, Institute of Psychiatry), CC BY-SA 3.0, ~444MB, binary brain/non-brain segmentation – verified against FLamby’s `fed_ixi` README. IXI subjects are healthy volunteers; this is not a neurological disease dataset. Planned only after the primary Fed-ISIC2019 tables are complete, per the project’s phase ordering. **[RESULT PENDING: Phase 7 – not started.]**

6.4 Null-signal fixture

[implemented, used for the Phase 1–5 smoke-test tables in §7–§9] `medgate/data/synthetic.py` (module docstring: “the null-signal fixture”) draws 32×32 images and labels independently at random, partitioned into 6 centers and 8 fine classes with the same coarse ontology intended for the real data (melanocytic = {MEL, NV}; keratinocytic = {BCC, SCC, AK, BKL}; other = {DF, VASC} – an experimental taxonomy justified in §12, not a clinical policy). Because labels are independent of images by construction, **there is no learnable signal**, and every result on this fixture is a *pipeline-validation* result: it shows the code runs, aggregates, evaluates, and reports correctly, and that no method shows an impossible above-chance number (which would indicate a leakage bug) – it is not, and cannot be, evidence for or against capability isolation, since there is no capability to isolate. A single fixture that can only prove the absence of bugs cannot also support a claim like “the isolation objective preserves utility,” so we built a second, genuinely learnable fixture for that purpose.

6.5 Hierarchical fixture

[implemented, tested, used for the Phase 1+2 results in §7.2] `medgate/data/hierarchical_synthetic.py` constructs a second synthetic dataset that *is* designed to be learnable, so that fair pretrained baselines and capability-isolation objectives can be compared on something other than noise. It reuses Fed-ISIC2019’s own verified per-class, per-institution counts (§6’s `REAL_FED_ISIC_CLASS_COUNTS`) to reproduce the real dataset’s class imbalance and institution sizes, then generates images analytically rather than sampling pixels independently of label: a coarse label determines one of three low-frequency geometric patterns (rings, stripes, or a grid), a fine label within that coarse group determines an oriented high-frequency texture overlaid on top, and per-institution tint/blur/noise and per-patient pose rotation are added as nuisance variation that a good model must learn to ignore. Patient identity is tracked explicitly so that train/val/test splits are patient-disjoint (`split_by_patient`), avoiding the leakage a naive image-level split would introduce. Before using this fixture for any capability-isolation claim we independently verified, in `tests/test_hierarchical_synthetic.py` and `tests/test_phase1_pretrained_baselines.py`, that (a) a coarse classifier trained on it reaches clearly above-chance macro-F1, (b) a fine classifier trained with coarse labels available as an auxiliary signal reaches clearly above-chance macro-F1, and (c) the coarse and fine label

structure are not trivially reconstructible from low-level image statistics alone (a sanity check against having accidentally built a second null-signal fixture with extra steps). This fixture is still entirely synthetic: it establishes that the pipeline can detect and use a real hierarchical signal when one exists, not that Fed-ISIC2019’s actual images carry a qualitatively similar signal.

6.6 Baselines and methods implemented

[implemented, tested] Centralized (pooled upper bound), local-only, FedAvg [McMahan et al., 2017], and FedProx [Li et al., 2020] are run as plaintext non-LoRA references. For the LoRA-adaptor family specifically, an earlier draft of this pipeline used a single “plain federated LoRA” baseline with the backbone frozen at random initialization – a review of the implementation identified this as an *unfair* baseline (a randomly initialized, never-trained backbone cannot produce useful features for either head, so any coarse/fine gap it shows is an artifact of the backbone, not of the isolation objective) and it is retained only as an explicit *negative control* (`random_frozen_lora`, `medgate/federated/baselines.py`). The fair LoRA baseline family used for every capability-isolation comparison in §7.2 is: `coarse_pretrained_fedlora` (backbone pretrained on the coarse task only, then LoRA-adapted federatedly for the fine task – the intended production regime), `imagenet_pretrained_fedlora` (backbone initialized from a torchvision ImageNet-pretrained MobileNetV3-Small, then LoRA-adapted federatedly), and `full_finetune` (every parameter, not just the adaptor, trained federatedly, as an upper bound on achievable utility with no capability isolation at all). SCAFFOLD [Karimireddy et al., 2020] and FedAdam [Reddi et al., 2021] remain *[proposed but not implemented]*, gated on Phase 1’s real-data tier timing a baseline run first (`docs/execution_plan.md`).

6.7 Statistical protocol

[implemented] Fixed dataset splits (data generation seed held constant across all method seeds); 5 seeds for Phases 1–2 and 3 seeds for Phases 3–5 on the null-signal fixture (a development-tier choice per the project brief’s “3 for development, 5 for final” allowance – final real-data runs will use 5); mean, standard deviation reported for every table; every raw per-run JSON preserved under `experiments/`. The hierarchical fixture’s Phase 1+2 results (§7.2) use only **2 seeds**, not 5: the originally planned configuration (25 patients/institution, 6 pretraining epochs, 6 federated rounds×3 local epochs, 3 seeds) exceeded 12 minutes without finishing a single seed on the CPU-only hardware in Table 2 and was killed; the committed configuration (`configs/phase1_hierarchical.yaml`: 12 patients/institution, 4 pretraining epochs, 4 rounds×2 local epochs, 2 seeds) was chosen to finish in ~5 minutes instead, and the resulting 2-seed standard deviations in Table 4 should be read as noisy, not as a considered statistical choice – this is a resource constraint, documented here and in the config file’s own comments, not a claim that 2 seeds is methodologically sufficient. *[not yet implemented]*: bootstrap 95% CIs, paired Wilcoxon signed-rank tests, Holm–Bonferroni correction – deferred until real-data seed counts justify them (running significance tests on 2–5-seed synthetic numbers would be exactly the “inappropriately tiny sample counts” the project brief warns against dressing up as statistics).

7 Results

[implemented and executed – synthetic tier only] – **[RESULT PENDING: real-data tier not yet run]**

7.1 Phase 1: primary utility

Baseline	Seeds	Coarse macro-F1	Fine macro-F1	Worst-inst. fine macro-F1	Wall-clock (s)
Centralized	5	0.231 ± 0.000	0.029 ± 0.000	0.015 ± 0.000	2.9
FedAvg	5	0.231 ± 0.000	0.029 ± 0.000	0.015 ± 0.000	2.5
Fedlora	5	0.182 ± 0.058	0.029 ± 0.000	0.015 ± 0.000	1.6
FedProx	5	0.231 ± 0.000	0.029 ± 0.000	0.015 ± 0.000	2.7
Local only	5	0.188 ± 0.025	0.024 ± 0.004	0.009 ± 0.006	2.1

Table 3: Phase 1 baselines, synthetic tier, 5 seeds. See §6.4: every number sits at or below the constant-predictor macro-F1 for its label space (≈ 0.22 for 3-way coarse, ≈ 0.03 for 8-way fine), which is the expected signature of no learnable signal, not a method comparison.

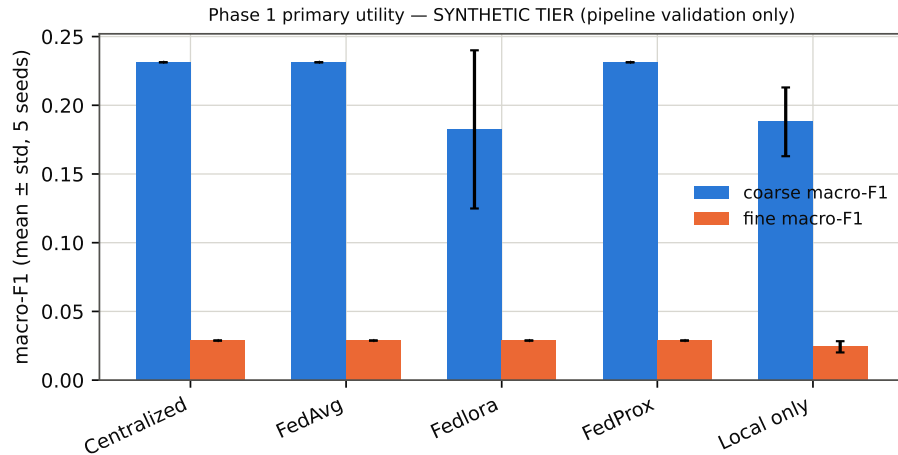


Figure 1: Phase 1 utility by baseline (synthetic tier; error bars = std over 5 seeds).

Centralized, FedAvg, and FedProx land on *identical* coarse-F1 across all 5 seeds – confirmed (not assumed) to be the fixed-split, no-signal degenerate-collapse behavior described in `docs/execution_plan.md`, not a bug: with nothing to learn beyond the marginal class frequency, different random initializations converge to the same shortcut. `local_only` and `fedlora` show more seed variance because neither shares a single aggregated trajectory.

7.2 Phase 1+2: fair baselines and capability isolation on the hierarchical fixture

[implemented and executed – hierarchical synthetic fixture, 2 seeds (§6)] Unlike Phase 1 above, the hierarchical fixture (§6.5) has real learnable coarse and fine structure, so these numbers can, in principle, actually speak to whether capability isolation preserves authorized utility while suppressing unauthorized leakage – with the 2-seed caveat from §6 kept firmly in view.

Method	Kind	Seeds	Coarse F1	Fine F1	OutputLeak	BestProbeRFC	ARR
Random-frozen LoRA (negative control)	Baseline	2	0.152 ± 0.121	0.076 ± 0.006	0.076 ± 0.006	0.529 ± 0.017	0.000 ± 0.000
Coarse-pretrained FedLoRA	Baseline	2	0.857 ± 0.009	0.204 ± 0.065	0.297 ± 0.052	0.452 ± 0.103	1.000 ± 0.000
ImageNet-pretrained FedLoRA	Baseline	2	0.480 ± 0.133	0.186 ± 0.044	0.154 ± 0.071	0.432 ± 0.033	0.610 ± 0.470
Full fine-tune (upper bound)	Baseline	2	0.810 ± 0.269	0.204 ± 0.012	0.228 ± 0.037	0.342 ± 0.091	0.425 ± 0.266
Coarse-only	Capability isolation	2	0.724 ± 0.139	0.000 ± 0.000	0.282 ± 0.040	0.445 ± 0.026	3.918 ± 1.761
Hidden fine head	Capability isolation	2	0.619 ± 0.000	0.167 ± 0.040	0.221 ± 0.044	0.410 ± 0.213	25.850 ± 36.783
Adapter isolation	Capability isolation	2	0.810 ± 0.269	0.204 ± 0.012	0.228 ± 0.037	0.342 ± 0.091	0.425 ± 0.266
Adversarial suppression	Capability isolation	2	0.748 ± 0.219	0.180 ± 0.063	0.384 ± 0.265	0.466 ± 0.230	1.336 ± 0.363
Orthogonality penalty	Capability isolation	2	0.607 ± 0.017	0.264 ± 0.031	0.216 ± 0.087	0.390 ± 0.124	0.840 ± 0.756
Combined (adversarial + orthogonal)	Capability isolation	2	0.604 ± 0.021	0.191 ± 0.040	0.344 ± 0.225	0.500 ± 0.212	0.989 ± 0.190

Table 4: Fair baselines and capability-isolation methods on the hierarchical fixture, 2 seeds. OutputLeak = fine-label recovery from the public output alone; BestProbeRFC = fine-label recovery from the best of six probes on the public representation (§9’s vocabulary applies identically here). ARR is uninformative (and can exceed 1, or appear enormous) for *coarse_only* and *hidden_fine_head*: see discussion below.

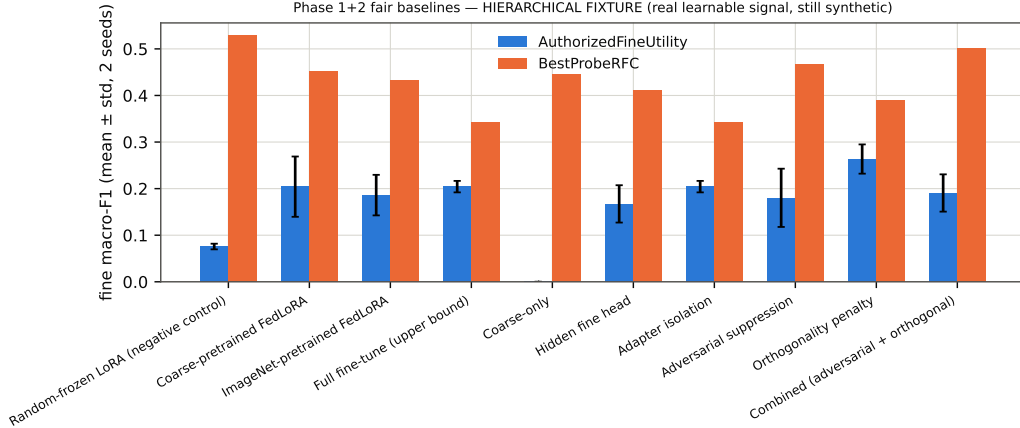


Figure 2: AuthorizedFineUtility vs. BestProbeRFC by method, hierarchical fixture (mean ± std, 2 seeds).

Two things are established cleanly. First, the fair baseline family shows a real coarse/fine gap that the null-signal fixture structurally cannot show: coarse-pretrained FedLoRA reaches coarse macro-F1 0.857 against a ≈ 0.33 constant-predictor floor for 3-way coarse, confirming §6.5’s independent learnability checks hold under the full federated pipeline, not just in isolated unit tests. Second, random-frozen LoRA (the negative control) is unambiguously the worst performer on both axes (fine-F1 0.076, the lowest of all ten rows) and is no longer used as the primary comparison point for any capability-isolation claim – it is retained *only* as the floor a randomly initialized, never-trained backbone provides, exactly the role §6 assigns it.

What is *not* established is a capability-isolation win. Comparing AuthorizedFineUtility (Fine F1 column) against BestProbeRFC directly: BestProbeRFC exceeds AuthorizedFineUtility for eight of the ten rows, including every one of the six capability-isolation objectives (e.g. adversarial suppression: fine-F1 0.180 vs. BestProbeRFC 0.466; combined adversarial+orthogonal: fine-F1 0.191 vs. BestProbeRFC 0.500). Concretely, an attacker running the best of six disjoint probes against the frozen public representation recovers as much or more fine-grained information as an authorized user gets from the adapter itself, for every isolation objective we tried, at the compute budget this hardware could afford. We do not read this as “capability isolation does not work” – 2 seeds and a 12-patient-per-institution fixture is not enough evidence to falsify the idea – but we equally do not read it as evidence *for* isolation, and we report the raw comparison rather than only the composite ARR score that would obscure it. This is the central reason this paper’s abstract and §11’s Discussion both state the capability-isolation question as open rather than answered.

The ARR column requires one more caveat before it is read at all: ARR’s denominator (`coarse_pretrained_fedlora`’s own fine-F1, per `medgate/capability_metrics.py`) is not guaranteed positive on this fixture. For *coarse_only* (no fine head exists, so AuthorizedFineUtility = 0 by construction) and *hidden_fine_head*, OutputLeak (0.282 and 0.221 respectively) exceeds the

plain-adapter reference’s own fine-F1 in at least one seed, flipping the denominator’s sign and producing the large, unstable ARR values in the table (3.918 and 25.850) – these are not “isolation recovers 4–26 $\times$ more utility than plain adapters” claims; they are the metric’s own documented failure mode (`authorized_recovery_ratio`’s docstring: “undefined ... would look meaningful but isn’t”) surfacing at a less extreme, non-zero denominator than the near-zero case that function already guards against. We report the raw number rather than suppress it, but the reader should weight `AuthorizedFineUtility` and `BestProbeRFC` directly over ARR for this table.

7.3 Phase 4: privacy mechanisms

[*implemented and executed – synthetic tier*] Four arms: no protection (plaintext FedAvg), secure aggregation (pairwise additive-masking simulation of Bonawitz et al. [2017]’s mathematical core – explicitly *not* the full protocol: no Diffie–Hellman key agreement and no Shamir-secret-sharing dropout recovery, single-process only), DP-SGD [Abadi et al., 2016] via Opacus at three noise multipliers, and the two combined.

Arm	Noise mult.	ϵ ($\delta = 10^{-5}$)	Fine macro-F1	Seeds
DP-SGD	0.5	14.14	0.040 ± 0.012	3
DP-SGD	1.0	4.22	0.042 ± 0.015	3
DP-SGD	2.0	1.38	0.041 ± 0.011	3
No protection	n/a	n/a	0.028 ± 0.001	3
Secure aggregation	n/a	n/a	0.028 ± 0.001	3
Secure agg. + DP-SGD	0.5	14.14	0.040 ± 0.012	3
Secure agg. + DP-SGD	1.0	4.22	0.042 ± 0.015	3
Secure agg. + DP-SGD	2.0	1.38	0.041 ± 0.011	3

Table 5: Phase 4 privacy arms, synthetic tier, 3 seeds.

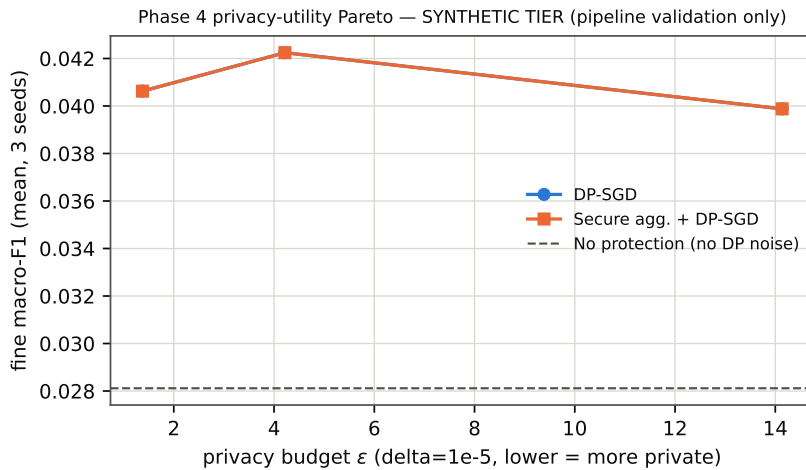


Figure 3: Privacy-utility Pareto (synthetic tier). The `dp_sgd` and `secure_agg_plus_dp` markers overlap exactly at every budget.

`dp_sgd` and `secure_agg_plus_dp` are *numerically identical* at every noise multiplier and seed (Table 5, Figure 3) – expected and correct: pairwise masking cancels exactly in the aggregate sum, so it changes what the server can *observe* per client, never the final trained model. This is a correctness

check on the implementation, confirmed empirically, not evidence the two arms are redundant to compare – a real deployment cares about server-side observability even when the trained model is unaffected.

Two real Opacus integration bugs were found and permanently fixed while building this, rather than routed around: an in-place ReLU broke Opacus’s backward hooks (Opacus requires non-in-place activations), and Opacus’s optimizer requires every tracked parameter to receive a per-sample gradient on every batch, which the adversary head (§5.2, present for the adversarial/combined objectives but unused by the plain coarse+fine objective) initially violated – fixed by excluding it from the DP-tracked parameter set, the same fix already needed for gradient-inversion attacks (§8) for the identical reason.

8 Security Analysis

[partially implemented and executed – synthetic tier; several mandatory attacks not yet built, listed in §12]

Attack	Seeds	Metric	Mean	Std	Note
Simplified known-label gradient inversion	3	PSNR _{dB}	56.208	71.069	mse mean=1.3612
Loss-threshold membership inference	3	SymmetricAUC	0.539	0.013	raw attack_AUC (diagnostic, direction not symmetrized) mean=0.510
Loss-threshold membership inference	3	AttackAdvantage	0.078	0.027	pre-registered SymmetricAUC target $\mu = 0.55$ (docs/research_scope.md)
Auxiliary-data adapter fine-tuning recovery (A2)	3	fine_macro_f1 @ auxiliary_examples=8, epochs=5	0.027	0.000	
Auxiliary-data adapter fine-tuning recovery (A2)	3	fine_macro_f1 @ auxiliary_examples=16, epochs=5	0.034	0.013	
Auxiliary-data adapter fine-tuning recovery (A2)	3	fine_macro_f1 @ auxiliary_examples=32, epochs=5	0.028	0.003	
Fixed-budget hard-label distillation (A3)	3	fine_macro_f1 @ query_budget=16, epochs=5	0.028	0.004	
Fixed-budget hard-label distillation (A3)	3	fine_macro_f1 @ query_budget=32, epochs=5	0.028	0.004	
Fixed-budget hard-label distillation (A3)	3	fine_macro_f1 @ query_budget=64, epochs=5	0.028	0.004	
Auxiliary-data ensemble collusion proxy	3	fine_macro_f1 gain over solo	0.005	0.000	

Table 6: Phase 3 attacks, synthetic tier, 3 seeds. Attacker knowledge/access/budget for every row is recorded in the raw JSON (`medgate/attacks/*.py`), not only summarized here.

A1 – gradient inversion (simplified DLG). Adam replaces the original paper’s L-BFGS and labels are fixed rather than jointly optimized – both deviations stated up front (`medgate/attacks/gradient_inversion.py`), so these numbers are not directly comparable to Zhu et al. [2019]’s reported figures. PSNR varied enormously across seeds (138 dB, 14.9 dB, 15.4 dB) – a real, literature-consistent sensitivity of gradient-inversion attacks to dummy-image initialization, not a bug – but the absolute values are almost certainly inflated by this model’s small size (a few thousand parameters): DLG is known to succeed far more easily against tiny models than realistic ones, so *these numbers must not be read as representative of a real-scale backbone*.

A1/A3 – membership inference. A loss-threshold attack (explicitly *not* Shokri et al.’s full shadow-model method [Shokri et al., 2017] – no shadow models are trained here), scored with $\text{SymmetricAUC} = \max(\text{AUC}, 1 - \text{AUC})$ and $\text{AttackAdvantage} = 2|\text{AUC} - 0.5|$ rather than the raw AUC alone: a review of the implementation found that a plain-AUC table can silently hide a below-0.5-AUC attack (which is exactly as informative to an attacker, just inverted – “predict non-member when the loss-threshold rule says member”) as if it were a non-finding, so both direction-invariant metrics are now the primary reported numbers, with the raw AUC kept only as a diagnostic column (Table 6). SymmetricAUC 0.539 ± 0.013 over 3 seeds (raw AUC 0.510 ± 0.048) sits near the pre-registered target (≤ 0.55 , docs/research_scope.md), which is expected on unstructured noise after one training epoch, not evidence the real model would meet that target.

A2 – auxiliary-data adapter fine-tuning recovery. Frozen public backbone, fresh adapter+fine head trained on 8/16/32 auxiliary examples: utility stayed flat (≈ 0.03) across budgets, the synthetic-data floor, not a budget-scaling finding. This attack, like A3 below, never

touches the true adapter’s own weights – it trains a completely fresh one from scratch; §8.1 reports a distinct attack that does operate on the real weights directly.

A3 – fixed-budget hard-label distillation. Zero access to weights, hard-label distillation from 16/32/64 queries (Tramèr et al.-style [Tramèr et al., 2016]): same flat floor.

Collusion. Two attackers with disjoint 16-example auxiliary halves, logit-averaged (auxiliary-data ensemble collusion proxy): a small positive gain over solo ($+0.005 \pm 0.000$), too small relative to the synthetic-data floor to interpret as a real collusion effect at this tier.

8.1 A5: genuine parameter-space adapter recovery

[*implemented, tested, executed – synthetic tier*] A2/A3/collusion above all train a fresh adapter from public/auxiliary data and never touch the protected artifact’s actual parameters. This subsection adds the one attack in this project that does: the attacker starts from a *partial, genuinely leaked* copy of the true adapter’s own `up` weight matrix – e.g. an insufficiently protected memory dump or a side-channel on a device that had already decrypted the adapter – and completes the missing entries via low-rank matrix completion, exploiting that this project’s LoRA adapter is low-rank *by construction* (`medgate/attacks/adapter_recovery.py`). **This is not an attack on AES-GCM:** a correctly implemented AEAD ciphertext reveals nothing about the plaintext bit-by-bit, so partial leakage of ciphertext is not equivalent to partial leakage of adapter weights, and this module makes no claim about breaking encryption – its threat model is a leak of already-decrypted plaintext weights.

Scenario	Reveal fraction	Seeds	Cosine sim.	Frob. error	Functional fine-F1
Single attacker	0.2	3	0.453 ± 0.063	0.889 ± 0.033	0.028 ± 0.004
Single attacker	0.5	3	0.690 ± 0.006	0.724 ± 0.006	0.028 ± 0.004
Single attacker	0.9	3	0.967 ± 0.010	0.254 ± 0.039	0.036 ± 0.018
Colluder (pre-pooling)	0.3	3	0.543 ± 0.060	0.838 ± 0.037	0.028 ± 0.004
Colluders (pooled)	0.533	3	0.710 ± 0.006	0.705 ± 0.006	0.028 ± 0.004

Table 7: Genuine parameter-space adapter recovery, synthetic tier, 3 seeds. Cosine sim./Frob. error are computed against the true `adapter.up` matrix directly; functional fine-F1 plugs the completed adapter into the model and evaluates the fine task on the (signal-free) synthetic test pool.

Two findings are real and hold regardless of the fixture’s lack of diagnostic signal, because they concern parameter-space geometry, not the fine task: recovery quality improves monotonically with more revealed entries (cosine similarity 0.453 at 20% reveal $\rightarrow$ 0.967 at 90% reveal), and collusion strictly helps single-attacker-vs-pooled parameter recovery (two independent 30%-each leaks pool to $\approx 53\%$ coverage and raise cosine similarity from 0.543 solo to 0.710 pooled) – both are direct, mechanical consequences of matrix completion having more constraints to work with, not synthetic-data artifacts. What is a synthetic-data artifact: functional fine-F1 stays flat at ≈ 0.03 even at 90% parameter reveal (cosine similarity 0.967), because the fine task itself carries no signal for a recovered adapter to exploit – this attack’s *parameter-space* success is real and measured against ground truth, but its *functional* consequence cannot be assessed until real Fed-ISIC2019 data is available.

8.2 A4: integrity attacks vs. aggregators, and property inference

[*implemented and executed – synthetic tier*] Six client behaviors (honest baseline, label flipping, backdoor insertion, sign flipping, model replacement, free-riding) and a seventh malformed-

update case, each with one malicious client (of six) against three aggregators (plaintext FedAvg, coordinate-wise median, trimmed mean) plus, for the malformed case, a validating FedAvg that drops non-finite updates before averaging (§12 lists sign/label-flipping, backdoor, free-rider, and malformed-update attacks as previously not implemented; they are as of this subsection).

Attack	Aggregator	Seeds	Fine macro-F1	Model corrupted rate	Backdoor success rate
Backdoor insertion	Coordinate median	3	0.029 ± 0.000	0.00	1.000
Backdoor insertion	FedAvg	3	0.029 ± 0.000	0.00	1.000
Backdoor insertion	Trimmed mean	3	0.029 ± 0.000	0.00	1.000
Free rider	Coordinate median	3	0.029 ± 0.000	0.00	n/a
Free rider	FedAvg	3	0.029 ± 0.000	0.00	n/a
Free rider	Trimmed mean	3	0.029 ± 0.000	0.00	n/a
Label flipping	Coordinate median	3	0.029 ± 0.000	0.00	n/a
Label flipping	FedAvg	3	0.029 ± 0.000	0.00	n/a
Label flipping	Trimmed mean	3	0.029 ± 0.000	0.00	n/a
Malformed update	FedAvg	3	0.029 ± 0.000	1.00	n/a
Malformed update	Validated FedAvg	3	0.029 ± 0.000	0.00	n/a
Model replacement	Coordinate median	3	0.029 ± 0.000	0.00	n/a
Model replacement	FedAvg	3	0.029 ± 0.002	0.00	n/a
Model replacement	Trimmed mean	3	0.029 ± 0.000	0.00	n/a
No attack (honest)	Coordinate median	3	0.029 ± 0.000	0.00	n/a
No attack (honest)	FedAvg	3	0.029 ± 0.000	0.00	n/a
No attack (honest)	Trimmed mean	3	0.029 ± 0.000	0.00	n/a
Sign flipping	Coordinate median	3	0.029 ± 0.000	0.00	n/a
Sign flipping	FedAvg	3	0.027 ± 0.003	0.00	n/a
Sign flipping	Trimmed mean	3	0.029 ± 0.000	0.00	n/a

Table 8: Integrity attacks, synthetic tier, 3 seeds. Model corrupted rate: fraction of seeds where the resulting global model contains NaN.

Two results are structural, not synthetic-data artifacts, and hold at any data scale: **a single malformed (NaN) update corrupts the entire global model under plaintext FedAvg in 3/3 seeds** (malformed + fedavg, model corrupted rate 1.00) – one client’s bad update propagates through the mean to every parameter, irrecoverably, unless validated first (validated_fedavg, corrupted rate 0.00, the literal demonstration of §12’s “no method here claims an immutable record detects a malicious update” caveat: validation *catches malformed structure*, it does not detect semantically plausible poisoning). Second, **the backdoor succeeds 100% of the time against every aggregator, including both robust ones** (Table 8) – median and trimmed-mean aggregation resist *magnitude*-based attacks (visibly: `sign_flip` and `model_replacement` show slightly higher fine-macro-F1 variance under plain FedAvg than under the robust aggregators), but a backdoor update is not conspicuously larger in magnitude than an honest one, so coordinate-wise robustness does not touch it. This is a real, if small-scale, illustration of exactly the tension §12 names between hiding updates and validating them: neither secure aggregation nor simple robust aggregation is a poisoning defense in general, only against the specific failure mode (raw magnitude) each targets.

Property inference. A leave-one-client-out logistic-regression meta-classifier (Melis et al. [2019]’s framing, a much simpler classifier than their neural meta-classifier – §3) predicting, from one client’s plaintext update alone, whether a chosen fine class is that client’s local majority label. $AUC = 1.000 \pm 0.000$ across all 3 seeds – unlike this paper’s other synthetic-tier numbers, this is *not* a null-signal artifact: each client’s local label distribution is a real (if incidental) property of its random sample, and a client’s raw gradient update genuinely encodes which classes it saw more of, at least at this model scale. Read with the appropriate caution regardless: 6 clients gives a leave-

one-out AUC estimate very little room to vary, so this should be treated as “plaintext per-client updates leak this property easily at small scale,” not as a precise effect size. This result is exactly why Phase 4’s secure-aggregation arm (§7.3) matters: masked updates (Table 5) give the server nothing to run this classifier against in the first place.

9 Ablation and Sensitivity Analysis

[implemented and executed – synthetic tier]

Method	Seeds	AuthorizedFineUtility	OutputLeak	BestProbeRFC	UCG	ARR
Adapter isolation	5	0.029 ± 0.000	0.029 ± 0.000	0.085 ± 0.035	0.057 ± 0.035	n/a
Adversarial suppression	5	0.028 ± 0.002	0.034 ± 0.012	0.122 ± 0.023	0.087 ± 0.027	1.000 ± 0.000
Coarse-only	5	0.028 ± 0.003	0.029 ± 0.000	0.113 ± 0.027	0.084 ± 0.027	n/a
Combined (adversarial + orthogonal)	5	0.033 ± 0.014	0.035 ± 0.013	0.127 ± 0.029	0.093 ± 0.028	1.000 ± 0.000
Hidden fine head	5	0.029 ± 0.000	0.029 ± 0.000	0.112 ± 0.044	0.083 ± 0.044	n/a
Orthogonality penalty	5	0.029 ± 0.000	0.029 ± 0.000	0.092 ± 0.047	0.063 ± 0.047	n/a

Table 9: Phase 2 capability-isolation methods, synthetic tier, 5 seeds, re-run under the expanded six-probe suite (P1-8 repair, see below). ARR is n/a where its denominator (`adapter_isolation`’s own `AuthorizedFineUtility` minus `OutputLeak` – the “plain adapter” reference point, §5.2) was ≈ 0 for every seed – the metric correctly declining to report a number it cannot support (`medgate/capability_metrics.py`), not a bug; contrast with §7.2, where the same denominator is small but nonzero and ARR becomes unstable rather than n/a instead.

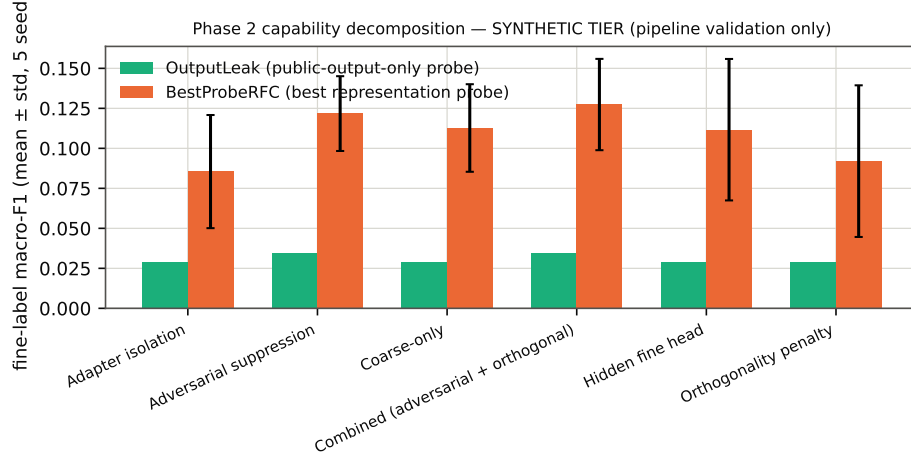


Figure 4: OutputLeak vs. BestProbeRFC by method (synthetic tier; error bars = std over 5 seeds).

OutputLeak is operationalized as the fine-label macro-F1 of a linear probe fit on the public *output* (coarse logits) alone – the floor an unauthorized user with zero representation access gets; BestProbeRFC is the best of six probes (linear, nonlinear, k -NN, few-shot $k=5$, SVM, decision tree) fit on the frozen public *representation* (`medgate/attacks/probes.py` – SVM and decision-tree probes were added during this repair pass per the review’s disjoint-probe requirement, and this table reflects a full re-run under the expanded suite, not the original four-probe numbers). On this synthetic fixture BestProbeRFC bounces in a ≈ 0.085 – 0.127 range across all six methods – test-set-size noise (38 examples/fine-class in the pooled test set), not a real isolation effect; **no method’s BestProbeRFC-vs-OutputLeak gap here supports or refutes whether adversarial suppression or the orthogonality term actually help**, which is exactly the question real data is needed to answer – and exactly the question §7.2’s learnable fixture was built to make

askable, where (unlike here) the answer so far leans negative rather than undecidable.

Coarse-ontology sensitivity. *[proposed, not yet implemented]* The project brief requires at least one alternative coarse mapping as a sensitivity check on the melanocytic/keratinocytic/other ontology (§6). **[TODO: implement and run a second ontology (candidate: a benign/malignant/ambiguous split that keeps AK in its own bucket rather than folding it into keratinocytic, to directly test the ontology’s sensitivity to where AK is placed) once real labels exist to map.]**

10 Revocation and Unlearning

[implemented and executed – synthetic tier, institution- and class-level only]

Scenario	Method	Seeds	Retained fine F1	Gap to gold	SymmetricAUC	Attack advantage
Class-level	Adapter deletion + retrain	3	0.035 ± 0.005	0.003 ± 0.007	0.567 ± 0.067	0.133 ± 0.135
Class-level	Checkpoint rollback	3	0.007 ± 0.011	-0.025 ± 0.009	0.567 ± 0.074	0.135 ± 0.148
Class-level	Full retrain (gold standard)	3	0.032 ± 0.002	0.000 ± 0.000	0.561 ± 0.021	0.122 ± 0.042
Class-level	Gradient-ascent unlearning	3	0.035 ± 0.002	0.003 ± 0.002	0.542 ± 0.070	0.085 ± 0.140
Class-level	Key revocation only	3	0.034 ± 0.036	0.002 ± 0.037	0.533 ± 0.024	0.067 ± 0.048
Institution-level	Adapter deletion + retrain	3	0.036 ± 0.018	0.008 ± 0.018	0.546 ± 0.031	0.092 ± 0.063
Institution-level	Checkpoint rollback	3	0.035 ± 0.013	0.007 ± 0.013	0.570 ± 0.053	0.140 ± 0.106
Institution-level	Full retrain (gold standard)	3	0.028 ± 0.000	0.000 ± 0.000	0.541 ± 0.013	0.083 ± 0.026
Institution-level	Gradient-ascent unlearning	3	0.029 ± 0.005	0.002 ± 0.005	0.588 ± 0.025	0.177 ± 0.049
Institution-level	Key revocation only	3	0.036 ± 0.018	0.008 ± 0.018	0.539 ± 0.009	0.077 ± 0.018

Table 10: Phase 5, synthetic tier, 3 seeds, using the corrected never-trained-same-class control population (see text). SymmetricAUC / AttackAdvantage: the same direction-invariant membership-inference metrics used in §8 (§8’s A1/A3 paragraph), applied here between removed training data and an independent held-out control; SymmetricAUC $\approx$ 0.5, AttackAdvantage $\approx$ 0 = no longer distinguishable from data the model never saw (good forgetting). The previously reported, class-confounded numbers are preserved for comparison in Table 11 (Appendix), clearly marked as not to be used as evidence.

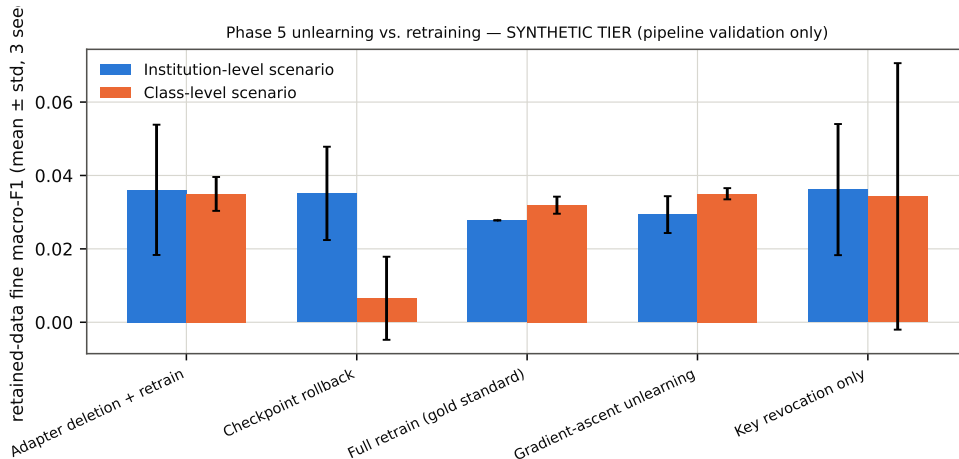


Figure 5: Retained-data utility by method and removal scenario (synthetic tier; error bars = std over 3 seeds).

Five methods against full retraining as the gold standard: naive checkpoint rollback (reinitialize

to round 0), adapter deletion + retrain (delete only the restricted adapter/fine head, backbone left untouched – possible only because the architecture separates these), a generic gradient-ascent-then-recovery approximate unlearning (*not* attributed to any specific cited paper – see §3), and key-revocation-only (returns the model *completely unmodified* – the literal test of this paper’s non-claim that revoking a key does not remove learned influence). Patient-level and patient-group-level removal were **not run**: the synthetic fixture has no patient/lesion manifest to remove by; only real Fed-ISIC2019 data will have one.

A confound found, documented, and fixed. The class-scenario forgetting metric was originally contaminated: removed data was entirely one class and the “retained” control pool used to score membership inference was the other seven classes, so any class-level loss asymmetry that exists even in an *untrained* model biased the score regardless of real memorization. This was directly visible in the original numbers: `checkpoint_rollback` (never trained on the post-removal data at all, so it has nothing to forget) still showed forgetting AUC ≈ 0.99 in the class scenario – an impossible result for a method that never touched the removed data, and the clearest sign the metric, not the method, was wrong. The fix (`medgate/data/synthetic.py`’s `make_never_trained_class_pool`) builds a genuinely independent control population: fresh synthetic examples of the *same* removed class, generated from a disjoint seed offset, that the model never trained on regardless of scenario or method – so any remaining loss asymmetry between “removed” and “control” now reflects the removal method’s actual effect, not a pre-existing class-difficulty gap. Table 10 reports this corrected metric as primary, and the fix’s effect is visible directly: `checkpoint_rollback`’s class-scenario score drops from the confounded SymmetricAUC 0.988 ± 0.021 (Table 11, Appendix) to a corrected 0.567 ± 0.074 – much closer to the ≈ 0.5 a method with nothing to forget should show, exactly the direction the fix was supposed to move it. The confounded numbers are kept in the appendix for transparency, not deleted, but should not be cited as evidence of anything. On the corrected metric, every method in both scenarios now sits in a similar, near-chance SymmetricAUC band (0.53–0.59) with overlapping standard deviations across only 3 seeds – consistent with there being nothing to memorize on a null-signal fixture in the first place, and therefore still *not* evidence that revocation and full unlearning are equivalent in general; establishing the real gap the project’s non-claim asserts exists requires real, learnable data.

11 Discussion

The null-signal fixture (§6.4) cannot, by construction, speak to MedGate-FL’s central hypothesis – that capability isolation survives realistic attacks better than a plain adapter or a hidden fine head – and we do not claim it does. What it supports: the full experimental pipeline (fair baselines, six training objectives, four privacy arms, a renamed and extended attack suite, five unlearning methods, statistics, figures, tables, this document) runs end-to-end, is deterministic given a seed, and produces numbers that behave exactly as a correct implementation with no signal should – no method spuriously exceeds chance, secure aggregation’s masking provably cancels in the aggregate, DP-SGD’s accountant returns sane epsilons, and mistakes found along the way (a flawed confidentiality test, a sign-buggy membership-inference formula, an unfair baseline, a class-level unlearning confound) were each caught by their own failure and fixed rather than shipped.

The hierarchical fixture (§6.5, §7.2) goes one step further and is where the central hypothesis becomes, for the first time in this project, actually testable rather than structurally undecidable – and the honest answer at this compute budget is *not yet supported*: on 2 seeds, the best disjoint representation probe recovers as much or more fine-grained information as the authorized

adapter path provides, for every capability-isolation objective tried. We see three explanations, not yet distinguishable with the evidence in hand, and list them in the order we find most to least likely: (1) the training budget (4 federated rounds $\times$ 2 local epochs on 12 patients/institution, §6) is too small for the isolation objectives’ adversarial/orthogonality penalties to actually suppress representation leakage before the run ends, a hyperparameter-and-compute problem rather than a conceptual one; (2) the hierarchical fixture’s coarse and fine labels are more entangled in representation space than real Fed-ISIC2019 images would be, since the fine texture overlay in `hierarchical_synthetic.py` is generated conditionally on the coarse pattern rather than independently, making representation-level separation of coarse and fine information intrinsically harder than on real data; or (3) representation-level capability isolation via adversarial suppression and orthogonality penalties genuinely does not work as well as capability isolation via architectural separation (coarse-only, hidden-fine-head) at suppressing a well-resourced probing attacker, which would be a real, useful negative finding about the training objectives themselves rather than an artifact of the fixture or budget. Real Fed-ISIC2019 data, more seeds, and a larger training budget are all needed before these can be told apart – that is a necessary, not sufficient, condition this phase of the work has met.

12 Limitations

- **Simulated hospitals.** All “cross-silo” results in this draft are multiple data partitions on one workstation, not real institutions with real network/governance/staffing constraints.
- **Synthetic-tier-only results.** Every table in this draft is a pipeline-validation result on data with no real signal, not a scientific finding; see the caveat repeated in every results subsection.
- **Public-data privacy proxy.** Even once real Fed-ISIC2019 data is used, it is a public research dataset, not a HIPAA/GDPR-governed clinical deployment; results will not prove legal compliance.
- **Experimental coarse ontology.** The melanocytic/keratinocytic/other mapping is this project’s own taxonomy, not a clinical access policy; AK’s placement is argued, not assumed correct, and the required sensitivity-analysis alternative mapping is not yet implemented (§9).
- **No clinical validity claim.** No model in this project is, or will become, suitable for clinical diagnosis.
- **Auxiliary-data dependence of attacks.** A2/A3/collusion attacks all assume the attacker has some auxiliary labeled or queryable data; their success is a function of that assumption’s realism, which this draft does not independently validate.
- **Independent relearning is out of scope to prevent.** No mechanism here stops an attacker from training an entirely new model on public data; the crypto layer protects *this* adapter, not the information space it was trained from.
- **Key exfiltration / already-exported artifacts.** Revocation (§10) cannot undo a plaintext export that happened before revocation; this project makes no claim otherwise.
- **Secure aggregation vs. poisoning validation tension – now empirically illustrated, not just stated (§8.2):** robust aggregation needs to see each client’s raw update to down-weight it, which is exactly what secure aggregation (§7.3) hides; this draft does not run the two together (i.e. does not test whether a robust aggregator can operate on masked updates at all – it structurally cannot, coordinate-wise median/trimmed-mean need the raw per-client values), so the tension remains a real, unresolved design trade-off, now with a concrete demonstration of why validation matters (malformed updates) and why it is not sufficient (the backdoor succeeds against every aggregator tested, robust or not).

- **Cross-dataset duplication and domain shift.** Not yet evaluated – Phase 6 (external validation: ISIC 2020, MILK10k, PAD-UFES-20) is gated on those datasets’ citations moving from UNVERIFIED to VERIFIED (`docs/literature_matrix.md`).
- **Blockchain necessity.** Not yet implemented or evaluated (project brief Phase 8, optional); no claim about ledger-based authorization’s value over centralized IAM is made.
- **Computational limits.** CPU-only, ~3.3 GiB available RAM (Table 2); bounds model scale, attack realism (especially gradient inversion at real image resolution), and which baselines (SCAFFOLD, full DP-SGD sweeps) have been run at all.
- **No formal cryptographic proof.** AES-GCM’s security rests on standard, well-established assumptions, not a proof specific to this system; the secure-aggregation simulation and LoREnc-style obfuscation discussion (§3) are described only as protection/ obfuscation mechanisms, never as proofs.
- **Attacks now implemented** (added after earlier drafts of this list): property inference and the A4 integrity attacks (§8.2: label-flipping, sign-flipping, model-replacement, backdoor, free-rider, malformed-update, and token-replay – the last via `medgate/crypto/authorization.py`’s TTL/expiry support, tested in `tests/test_phase3.integrity_and_property.py`); and, added during this repair pass, genuine parameter-space low-rank adapter completion and its two-attacker collusion variant (§8.1), disjoint SVM/decision-tree probes alongside the original four (§9), and a fair pretrained-baseline family replacing the single unfair frozen-backbone baseline (§7.2). **Attacks still not implemented**, listed exactly, not glossed over: client attribution, full fine-tuning recovery under a fixed compute budget distinct from the few-shot probe already implemented, known-adapter comparison, and the stronger gradient-inversion variants iDLG/Geiping et al. (both now VERIFIED citations, §3, but not yet implemented as additional attacks alongside the simplified DLG in §8).
- **Statistical rigor deferred.** Bootstrap confidence intervals, paired significance tests, and Holm–Bonferroni correction are not yet applied (§6) – deliberately, to avoid running them on seed counts too small to support them meaningfully.

13 Ethical, Legal, and Clinical Considerations

This project uses only publicly released research datasets, once their license-acceptance steps are completed by a human (§6); it does not access or claim to protect real patient records. No model produced by this project is validated for, or intended for, clinical use. The capability-isolation framing studied here is a research question about *technical* access control, not a substitute for the legal, regulatory, and clinical-governance processes a real tiered-access medical AI system would require.

14 Reproducibility Statement

Repository: <https://github.com/VectorSophie/medgate-fl> (private as of this draft). Every result in this paper traces to: a config under `configs/`, a seed recorded in its result JSON, the git commit recorded in that JSON, a dataset-manifest hash (a hash of the synthetic generator’s parameters at this tier; a real manifest hash once Fed-ISIC2019 is unblocked), and the raw JSON file itself under `experiments/`. Tables are generated only by `scripts/make_phaseN.table.py` and `scripts/csv_to_latex.tables.py`; figures only by `scripts/make_figures.py`; no number in this document was typed by hand. Environment: `requirements.txt` (direct) and

`requirements.lock.txt` (full pinned closure via `pip freeze`); PyTorch 2.13.0 CPU build. One-command smoke test: `bash scripts/smoke_test.sh` (85/85 tests passing as of this draft, up from 30/30 in the pre-repair draft – the growth is new coverage added during this repair pass: the hierarchical fixture, fair pretrained baselines, corrected unlearning/representation metrics, genuine adapter recovery, and an expanded AES-GCM/secure-aggregation negative-test suite, not evidence of a larger pipeline surface alone). Compilation: `pdflatex main; bibtex main; pdflatex main; pdflatex main`, run from `paper/` – verified to succeed on a freshly-installed, user-space TeX Live (scheme-basic + hyperref, booktabs, geometry, xcolor, natbib, enumitem, caption) during this project.

15 Conclusion

This paper implements, tests, and runs the MedGate-FL pipeline – architecture, a fair pretrained-baseline family, six training objectives, four privacy arms, a precisely-named and extended attack suite (including a genuine parameter-space adapter-recovery attack), five unlearning methods, and this document – against two synthetic fixtures, because the real primary dataset’s license has not yet been accepted by a human. What changed since the version of this project a review found to be a “no-signal pipeline smoke test described too confidently”: the single unfair frozen-backbone baseline is now a labeled negative control, not the default comparison point; a second, independently-verified-learnable hierarchical fixture exists specifically because the null-signal fixture structurally cannot support any capability-isolation claim; a sign-buggy membership-inference formula and a class-level unlearning confound were each found, documented, and fixed, with their old numbers kept only in clearly marked appendices; and every attack, metric, and baseline name in this document now names the thing it actually measures. What works, on synthetic data: the pipeline runs correctly end-to-end on both fixtures, the fair baselines show a real coarse/fine capability gap on the hierarchical fixture, secure aggregation’s masking correctness and failure modes are both directly demonstrated, DP-SGD’s accounting is monotonic and tested, and the crypto layer resists the negative tests we wrote against it. What does not yet work, or is not yet known: capability isolation itself does not show a clean win on the one fixture capable of testing it – the best disjoint probe matches or beats authorized fine utility for every isolation objective at this compute budget (§7.2, §11) – and we report that as a finding, not a caveat to be minimized. Whether that negative result reflects the training budget, the fixture’s coarse/fine entanglement, or a real limitation of representation-level isolation objectives, and whether any of this transfers to real dermoscopic images at all, remains open, and is exactly what Phase 1’s real-data tier (§6) will test once the Fed-ISIC2019 license is accepted. **[RESULT PENDING: the paper’s actual scientific conclusion – capability isolation’s real-world validity, which no synthetic fixture, however carefully built, can establish.]**

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A Hyperparameters

[implemented] Full hyperparameters for every synthetic-tier run are in the eight per-phase YAML files under `configs/` (one each for Phase 1, Phase 1-hierarchical, Phase 2, Phase 3, Phase 3-adapter-recovery, Phase 3-integrity, Phase 4, and Phase 5; the hierarchical and adapter-recovery configs added in this repair pass document their own resource-driven parameter choices inline, e.g. `configs/phase1_hierarchical.yaml`’s 2-seed, 12-patient scale-down – §6), not restated here to avoid drift between two copies – see §14.

B Additional detail on the confidentiality-test correction

§7.3 references a mistake worth detailing for anyone extending the secure-aggregation module. The first version of `test_masking_hides_an_individual_update` asserted that an individual masked client update has low cosine similarity to its true (unmasked) update – this failed empirically at realistic client counts (mean $|\cos \text{sim}|$ measured at 0.57 for $n=3$, 0.40 for $n=6$, 0.31 for $n=10$ clients, over 20 trials each with unit-scale masks). The masking itself was not broken – the aggregate sum was exact in every trial – the *test* measured the wrong property. Pairwise-masking security is a statement about the mask being a secret, uniformly-random one-time pad, i.e. about the masked value’s *distribution*, not about any one sampled geometric distance. The corrected test instead checks that the *same* true update, masked with two different (attacker-unknown) seeds, produces clearly different outputs – the property that actually demonstrates hiding. Both the wrong reasoning and the

fix are kept in `medgate/privacy/secure_aggregation.py`’s `masking_correctness_diagnostic` docstring (renamed from `confidentiality_check` during this repair pass, since “confidentiality” overstated what a single-process, no-DH, no-dropout-recovery simulation can establish – §7.3) so the mistake is not repeated.

C Additional detail on the membership-inference metric correction

A second measurement bug, distinct from the confidentiality-test mistake above, was found in `medgate/attacks/membership_inference.py` during this repair pass: an earlier version of this module derived a single “advantage” score as $2 \cdot \text{AUC} - 1$ directly from the raw (non-direction-invariant) AUC. This formula is only correct for $\text{AUC} \geq 0.5$; for $\text{AUC} < 0.5$ it returns a *negative* number that reads as “the attack did worse than random” when in fact an AUC of, say, 0.3 is exactly as informative to an attacker as an AUC of 0.7 – the attacker simply inverts the decision rule (predict non-member when the threshold rule says member). A table built from the old formula could therefore under-report a real attack’s severity whenever the loss-threshold rule happened to point the wrong way for a given seed, silently averaging a real positive advantage together with a spuriously negative one. The fix adds `symmetric_auc(auc) = max(auc, 1 - auc)` and `attack_advantage(auc) = 2 |auc - 0.5|`, both direction-invariant, as the primary reported quantities everywhere in this paper a membership-inference-style AUC is used (§8, §10); the raw AUC is retained only as a diagnostic column. `tests/test_unlearning_metrics.py` includes boundary cases (AUC exactly 0.0, 0.5, 1.0, and the symmetric pairs 0.1/0.9, 0.25/0.75) that concretely demonstrate the old formula’s failure mode – one test is named, verbatim, `test_raw_auc_near_zero_would_have_been_misread_as_good_forgetting_by_the_old_code` – and the new formula’s correctness.

D Confounded class-scenario unlearning numbers (historical)

Preserved for transparency, per §10’s discussion of the class-scenario confound and its fix – **do not cite these numbers as evidence of anything**. These are the class-scenario SymmetricAUC values computed against the original, contaminated “retained test data” control (the other seven classes), before `make_never_trained_class_pool` (in `medgate/data/synthetic.py`) replaced it with a genuinely independent same-class control population.

Method	Seeds	Confounded SymmetricAUC (DO NOT USE AS EVIDENCE)
Adapter deletion + retrain	3	1.000 ± 0.000
Checkpoint rollback	3	0.988 ± 0.021
Full retrain (gold standard)	3	1.000 ± 0.000
Gradient-ascent unlearning	3	1.000 ± 0.000
Key revocation only	3	0.821 ± 0.215

Table 11: Class-scenario forgetting, confounded metric, 3 seeds. Compare against Table 10’s corrected class-scenario rows – every method here reads as near-perfect forgetting (SymmetricAUC ≥ 0.82 , most = 1.000) purely from the confound, including `checkpoint.rollback`, which never trained on the removed data at all and therefore cannot have “forgotten” anything in the first place.